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1. It is given that

$$y = x^2 e^{ax}$$

where a is a constant.

Prove by mathematical induction that, for all integers $n \geq 2$,

$$\frac{d^n y}{dx^n} = a^{n-2} (a^2 x^2 + 2nax + n(n-1)) e^{ax} \quad [6]$$

2. Prove by mathematical induction that, for every positive integer n ,

$$\frac{d^n}{dx^n}(x^3 e^{-x}) = (-1)^n e^{-x} (x^3 - 3nx^2 + 3n(n-1)x - n(n-1)(n-2)) \quad [7]$$

3. Given that

$$y = e^{-2x} \cosh 3x$$

prove by induction that for $n \in \mathbb{N}$

$$\frac{d^n y}{dx^n} = e^{-2x} \left(\frac{1 + (-5)^n}{2} \cosh 3x + \frac{1 - (-5)^n}{2} \sinh 3x \right) \quad [6]$$

4. Let

$$y = x^2 \cosh x$$

Prove by induction that, for all integers $n \geq 1$,

$$\frac{d^{2n}y}{dx^{2n}} = x^2 \cosh x + 4nx \sinh x + 2n(2n-1) \cosh x$$

[6]

5. A function is defined by $y = f(t)$ where $f(t) = (1 + at) \ln(1 + at)$ and a is a constant.

(a) By considering $\frac{d^2y}{dt^2}$, $\frac{d^3y}{dt^3}$, $\frac{d^4y}{dt^4}$ and $\frac{d^5y}{dt^5}$, make a conjecture for a general formula for $\frac{d^ny}{dt^n}$ in terms of n , a and t for any integer $n \geq 2$. [3]

(b) Use induction to prove the formula conjectured in part (a). [4]

(c) In the case when $f(t) = (1 + 3t) \ln(1 + 3t)$, find the rate at which the 6th derivative of $f(t)$ is varying when $t = \frac{5}{3}$. [2]

6. The function $f(x)$ is defined by $f(x) = (1+x)^{-\frac{1}{2}}$, for $x > -1$.

(a) Prove by mathematical induction that the n th derivative of $f(x)$, $f^{(n)}(x)$, for all $n \geq 1$, is given by

$$f^{(n)}(x) = \frac{(-1)^n(2n)!}{2^{2n}n!(1+x)^{n+\frac{1}{2}}} \quad [4]$$

(b) Hence prove that

$$(1+x)^{-\frac{1}{2}} = 1 - \frac{x}{2} + \frac{3x^2}{8} - \frac{5x^3}{16} + \dots + (-1)^n \frac{(2n)!}{2^{2n}(n!)^2} x^n + \dots$$

$$\text{for } -1 < x \leq 1. \quad [3]$$

7. It is given that

$$y = (ax + 1)^2 \ln(ax + 1)$$

where a is a positive constant. Prove by mathematical induction that, for every integer $n \geq 3$,

$$\frac{d^n y}{dx^n} = 2(-1)^{n-3} \frac{(n-3)!a^n}{(ax+1)^{n-2}}$$

[6]

8. The function f is such that $f''(x) = -f(x)$.

Prove by mathematical induction that, for every positive integer n ,

$$\frac{d^{2n}}{dx^{2n}} (x^2 f(x)) = (-1)^n (x^2 f(x) - 4nx f'(x) - 2n(2n-1)f(x)) \quad [7]$$